

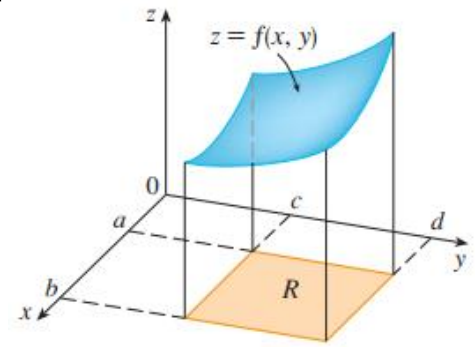
Unit-III
Multivariable Calculus-Integration
MA1002-1
(For I-B and I-D sections)

Double integration:

Consider a function $f(x, y)$ of two variables defined on a region of two dimensional plan say R , then a double integral of a function of two variables over a region R is

defined by,

$$\iint_R f(x, y) \, dA = \int_{x_1}^{x_2} \int_{y_1}^{y_2} f(x, y) \, dx \, dy.$$



Evaluation of double integration:

Case (i): When y_1 and y_2 are functions of x , (that is, $y_1 = f_1(x)$ and $y_2 = f_2(x)$), x_1 and x_2 are constants, $f(x, y)$ first integrated with respect to y keeping x fixed between the limits y_1, y_2 and the resulting expression is integrated w. r. t x within the limits x_1, x_2 that is,

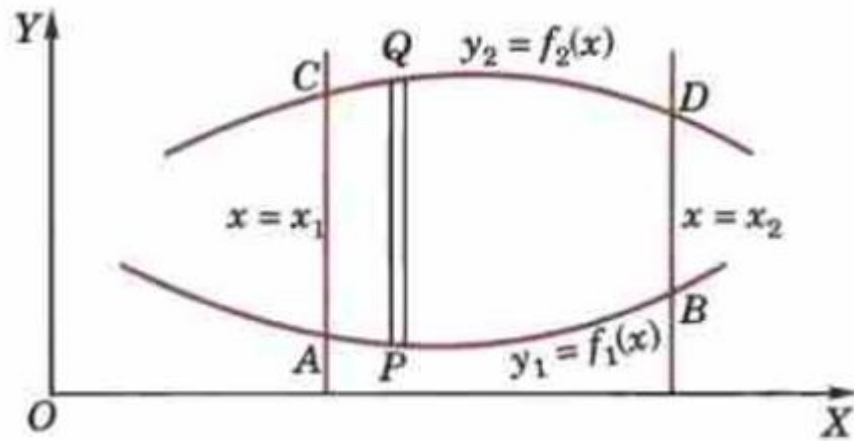
$$\int_{x_1}^{x_2} \left[\int_{y_1=f_1(x)}^{y_2=f_2(x)} f(x, y) \, dy \right] \, dx$$

where the integration is carried from the inner to the outer rectangle.

Figure 1 illustrates this process. Here AB and CD are the two curves whose equations are $y_1 = f_1(x)$ and $y_2 = f_2(x)$. PQ is a vertical strip of width dx .

Then the inner integral means that the integration is along one edge of the strip PQ from P to Q (x remaining constant), while the outer integral corresponds to the sliding of the edge from AC to BD .

Thus the whole region of integration is the area $ABCD$.



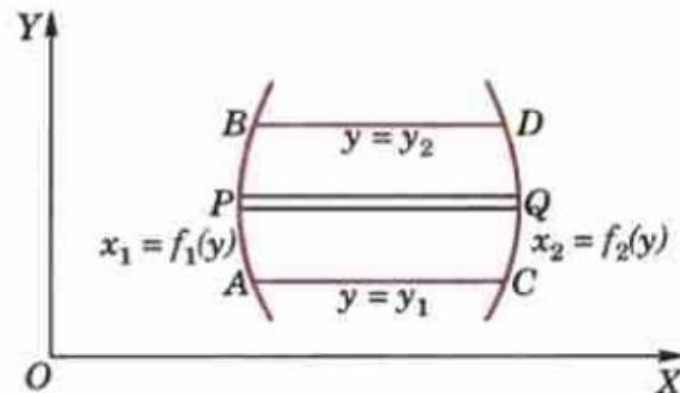
Case (ii): When x_1 and x_2 are functions of y , (that is, $x_1 = f_1(y)$ and $x_2 = f_2(y)$), y_1 and y_2 are constants, $f(x, y)$ first integrated with respect to x keeping y fixed between the limits x_1, x_2 and the resulting expression is integrated w. r. t x within the limits y_1, y_2 that is,

$$\int_{y_1}^{y_2} \int_{x_1=f_1(y)}^{x_2=f_2(y)} f(x, y) dx dy$$

Here AB and CD are the two curves whose equations are $x_1 = f_1(y)$ and $x_2 = f_2(y)$. PQ is a horizontal strip of width dy .

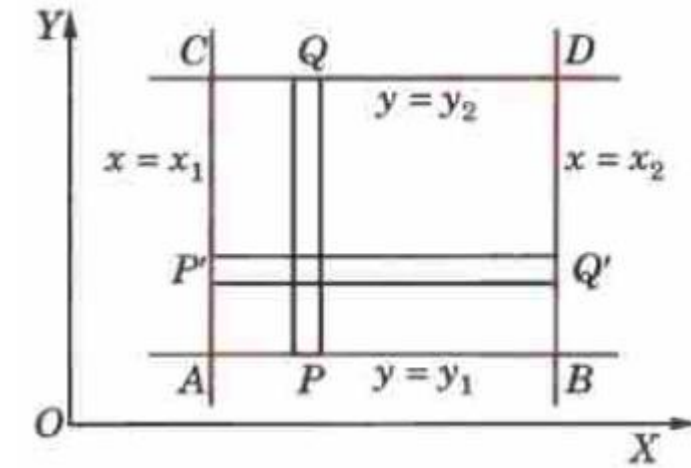
Then the inner integral means that the integration is along one edge of the strip PQ from P to Q (y remaining constant), while the outer integral corresponds to the sliding of the edge from AC to BD .

Thus the whole region of integration is the area $ABCD$.



Case(iii): When both pairs of limits are constants, the region of integration is the rectangle $ABCD$. Choose (or change) the order of integration (by looking at order of differentials) and use (i) or (ii) whichever is easier for integration.

$$\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y) dx dy$$



Thus for the constant limits, it hardly matters whether we first integrate w. r. t x and then w.r.t y or vice versa.

Problems:

Evaluate the following double integrals:

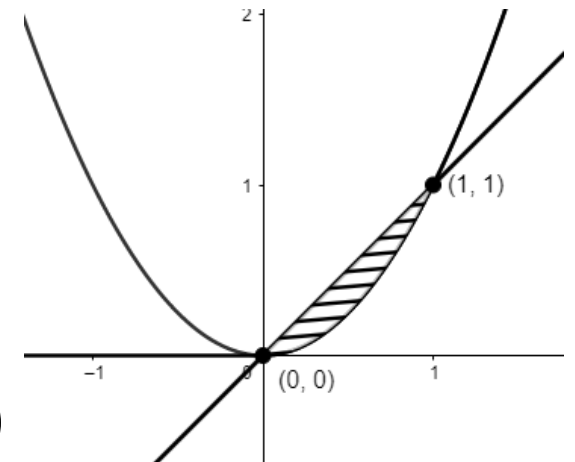
1. $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) \, dx \, dy.$

Solution: $I = \int_0^1 \left\{ \int_x^{\sqrt{x}} (x^2 + y^2) \, dy \right\} dx = \int_0^1 \left[x^2 y + \frac{y^3}{3} \right]_x^{\sqrt{x}} dx$

$$= \int_0^1 \left[x^2 \sqrt{x} + \frac{(\sqrt{x})^3}{3} - x^2(x) - \frac{x^3}{3} \right] dx = \int_0^1 \left[x^{\frac{5}{2}} + \frac{x^{\frac{3}{2}}}{3} - \frac{4x^3}{3} \right] dx$$

$$= \left[\frac{x^{\frac{7}{2}}}{\frac{7}{2}} + \frac{x^{\frac{5}{2}}}{3 \times \frac{5}{2}} - \frac{x^4}{3} \right]_0^1 = \frac{2}{7} + \frac{2}{15} - \frac{1}{3} = \frac{9}{105}.$$

(Region of integration for the given double integral is :)



2. $\int_0^1 \int_0^{x^2} e^{\frac{y}{x}} dx dy$.

Solution: $I = \int_0^1 \left\{ \int_0^{x^2} e^{\frac{y}{x}} dy \right\} dx$

$$= \int_0^1 \left[\frac{e^{\frac{y}{x}}}{\frac{1}{x}} \right]_0^{x^2} dx = \int_0^1 \left[x e^{\frac{x^2}{x}} - x e^0 \right] dx$$

$$= \int_0^1 [x e^x - x] dx = \left[\{x e^x - \int e^x dx\} - \frac{x^2}{2} \right]_0^1 = \left[\{x e^x - e^x\} - \frac{x^2}{2} \right]_0^1$$

$$= e - e - 0 + e^0 - \frac{1}{2} = \frac{1}{2}.$$

3. $\int_1^2 \int_0^{2-y} xy dx dy$.

Solution: $I = \int_1^2 \left\{ \int_0^{2-y} xy dx \right\} dy = \int_1^2 \left\{ \frac{x^2 y}{2} \right\}_0^{2-y} dy = \frac{1}{2} \int_1^2 \{y(2-y)^2 - 0\} dy$

$$\begin{aligned}
&= \frac{1}{2} \int_1^2 \{y(4 - 2y + y^2)\} dy &= \frac{1}{2} \int_1^2 \{4y - 2y^2 + y^3\} dy \\
&= \frac{1}{2} \left[\frac{4y^2}{2} - \frac{2y^3}{3} + \frac{y^4}{4} \right]_1^2 &= \frac{1}{2} \left[2(2^2) - \frac{2}{3}(2^3) + \frac{2^4}{4} - 2(1)^2 + \frac{2}{3}(1)^3 - \frac{1^4}{4} \right] \\
&= \frac{61}{12}.
\end{aligned}$$

4. $\int_0^2 \int_0^{x^2} x(x^2 + y^2) \, dy \, dx.$

Solution: $I = \int_0^2 \left\{ \int_0^{x^2} (x^3 + xy^2) \, dy \right\} dx$

$$= \int_0^2 \left[x^3 y + \frac{xy^3}{3} \right]_0^{x^2} dx = \int_0^2 \left[x^3(x^2) + \frac{x(x^2)^3}{3} - 0 \right] dx$$

$$= \int_0^2 \left[x^5 - \frac{x^7}{3} \right] dx = \left[\frac{x^6}{6} - \frac{x^8}{24} \right]_0^2 = \left[\frac{2^6}{6} - \frac{2^8}{24} \right] = 0.$$

5. $\int_1^4 \int_3^5 x^2 y \, dy \, dx$.

Solution: $I = \int_1^4 \left\{ \int_3^5 x^2 y \, dy \right\} dx$

$$= \int_1^4 \left[\frac{x^2 y^2}{2} \right]_3^5 dx = \frac{1}{2} \int_1^4 [x^2 (5)^2 - x^2 (3^2)] dx$$

$$= \frac{1}{2} \int_1^4 16x^2 \, dx = 8 \left[\frac{x^3}{3} \right]_1^4 = \left[\frac{4^3}{3} - \frac{1^3}{3} \right] = 8(21) = 168.$$

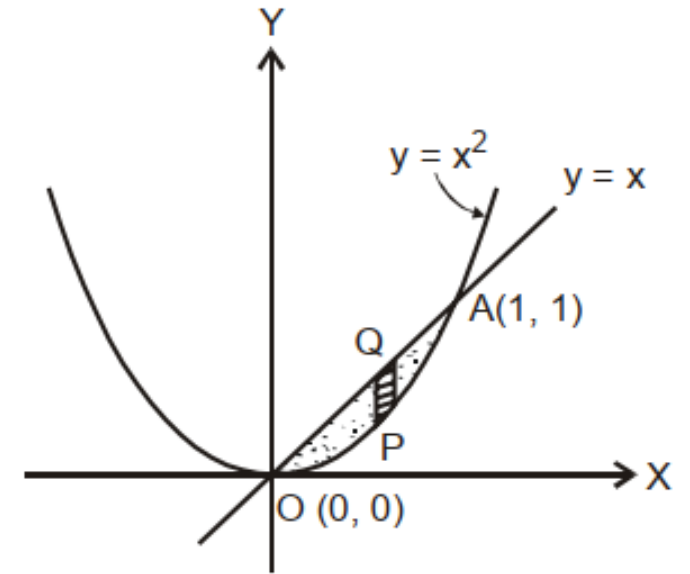
6. Evaluate $\iint_D xy(x+y) \, dx \, dy$, if D is the region bounded between $y = x^2$ and $y = x$.

Solution: We have $y = x^2$ and $y = x$ which implies $x^2 - x = 0$ i.e. either $x = 0$ or $x = 1$.

Further, if $x = 0$ then $y = 0$; if $x = 1$ then $y = 1$. Means the two curves intersect at points $(0, 0)$, $(1, 1)$.

∴ The region D of integration is dotted region in the figure.

Over the vertical strip PQ (say), if y changes as a function of x such that P lies on the curve $y = x^2$ and Q lies on the line $y = x$ and finally the strip slides between $x = 0$ to $x = 1$. Thus the region D can be expressed as:
 $0 \leq x \leq 1, x^2 \leq y \leq x$, (as shown in the figure).



$$\begin{aligned}
 \therefore \quad \iint_D xy(x+y) dx dy &= \int_0^1 \left(\int_{x^2}^x xy(x+y) dy \right) dx \\
 &= \int_0^1 \left\{ \left(x^2 \frac{y^2}{2} + x \frac{y^3}{3} \right)_{x^2}^x \right\} dx \\
 &= \int_0^1 \left\{ \left(\frac{x^4}{2} + \frac{x^4}{3} \right) - \left(\frac{x^6}{2} + \frac{x^7}{3} \right) \right\} dx \\
 &= \int_0^1 \left(\frac{5}{6} x^4 - \frac{1}{2} x^6 - \frac{1}{3} x^7 \right) dx \\
 &= \left[\frac{5}{6} \times \frac{x^5}{5} - \frac{1}{2} \frac{x^7}{7} - \frac{1}{3} \frac{x^8}{8} \right]_0^1 = \frac{1}{6} - \frac{1}{14} - \frac{1}{24} = \frac{3}{56}
 \end{aligned}$$

7. Evaluate $\iint_D (x^2 + y^2) dx dy$, if D is the region bounded between $x = 2, y = 1$ and $y = x^2$.

Solution: Here the region of integration is bounded is ABC.

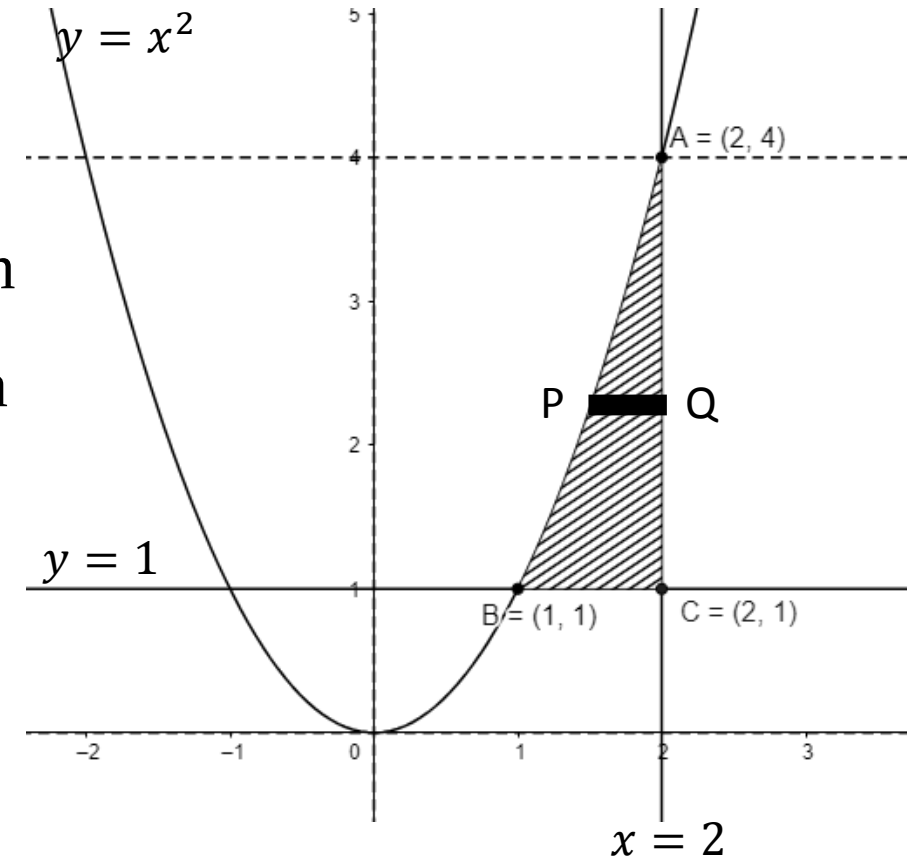
So, boundary points of the region are

$A = (2, 4), B = (1, 1)$ and $C = (2, 1)$ as shown in the figure. Over the horizontal strip PQ, x varies from the curve $x = \sqrt{y}$ to the line $x = 2$, and y varies from $y = 1$ to $y = 4$.

$$\iint_D (x^2 + y^2) dx dy = \int_1^4 \left\{ \int_{\sqrt{y}}^2 (x^2 + y^2) dx \right\} dy$$

$$= \int_1^4 \left[\frac{x^3}{3} + y^2 x \right]_{\sqrt{y}}^2 dy$$

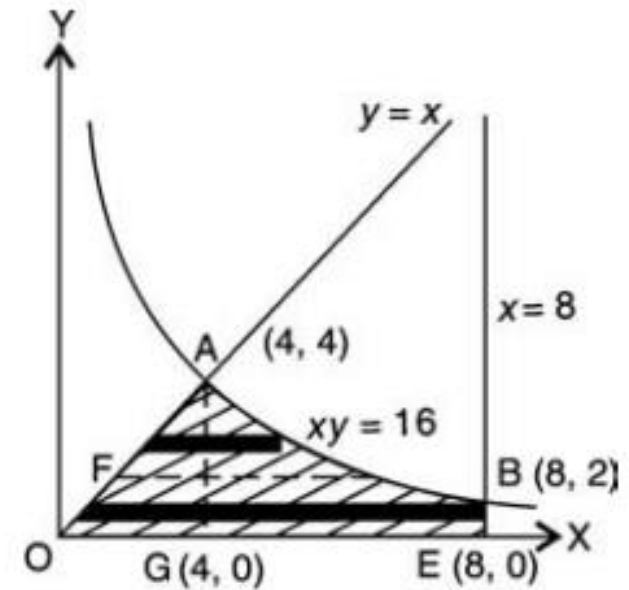
$$= \int_1^4 \left[\frac{(2)^3}{3} + y^2(2) - \frac{(\sqrt{y})^3}{3} - y^2\sqrt{y} \right] dy$$



$$= \int_1^4 \left[\frac{8}{3} + 2y^2 - \frac{y^{\frac{3}{2}}}{3} - y^{\frac{5}{2}} \right] dy$$

$$= \left[\frac{8y}{3} + \frac{2y^3}{3} - \frac{2y^{\frac{5}{2}}}{15} - \frac{2y^{\frac{7}{2}}}{7} \right]_1^4 = \frac{32}{3} + \frac{2(4^3)}{3} - \frac{2(4)^{\frac{5}{2}}}{15} - \frac{2(4)^{\frac{7}{2}}}{7} - \frac{8}{3} - \frac{2}{3} + \frac{2}{15} + \frac{2}{7}$$

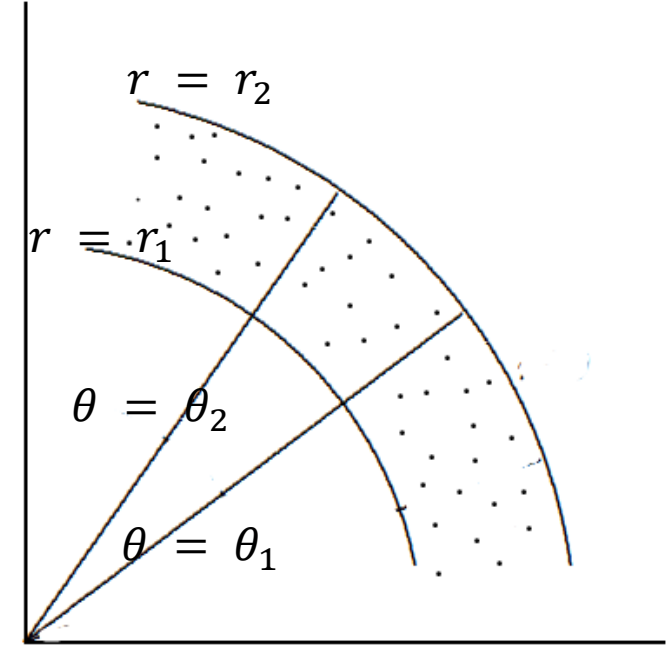
$$= \frac{1006}{105},$$



Evaluation of double integrals (polar form):

To evaluate the double integral $\int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r, \theta) \, dr \, d\theta$ over the region D bounded by the curve $r = r_1$, $r = r_2$ & the straight lines $\theta = \theta_1$, $\theta = \theta_2$.

We first integrate w. r. t. r (keeping θ constant) between the limits r_1 & r_2 & then integrating the resulting expression w.r.t. θ between the limits θ_1 & θ_2 .



1. Evaluate $\iint_D r \sin \theta \, dr \, d\theta$, if D is the region bounded by $r = a(1 - \cos \theta)$.

Solution:

The shaded region in Fig. below is the region of integration D . Here, r varies from 0 to $a(1 - \cos \theta)$ and θ varies from 0 to π (region is above the initial line).

Therefore,

$$I = \iint_D r \sin \theta \, dr \, d\theta = \int_0^\pi \int_0^{a(1-\cos \theta)} r \sin \theta \, dr \, d\theta$$

$$= \int_0^\pi \sin \theta \left(\frac{r^2}{2} \right)_0^{a(1-\cos \theta)} d\theta$$

$$= \int_0^\pi \sin \theta \left(\frac{a^2(1-\cos \theta)^2}{2} - 0 \right) d\theta$$

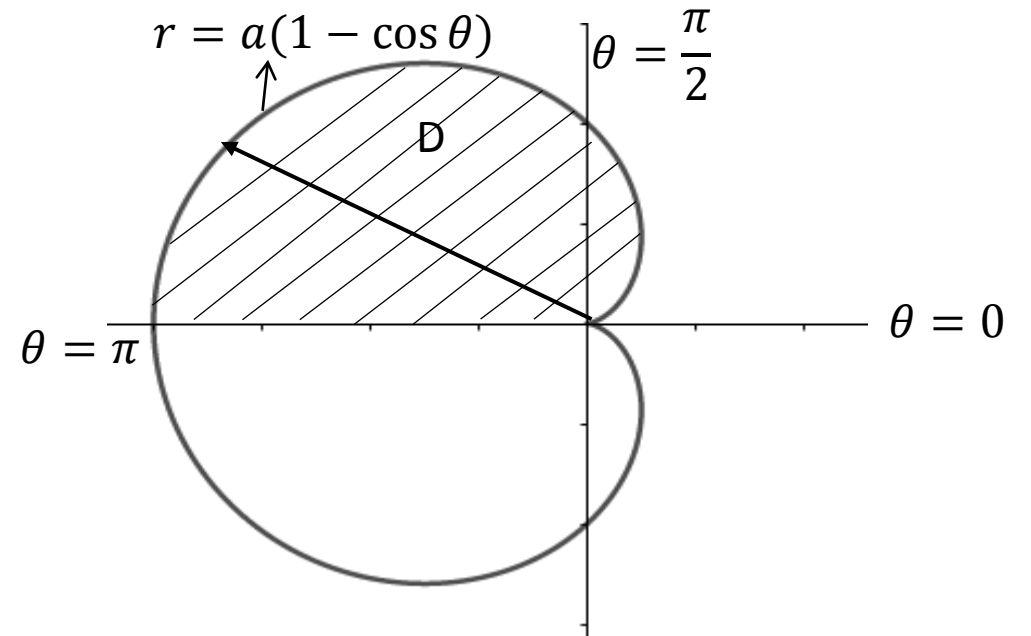
$$= \frac{a^2}{2} \int_0^\pi \sin \theta (1 - \cos \theta)^2 d\theta$$

Put $1 - \cos \theta = t$ then $\sin \theta \, d\theta = dt$

when $\theta = 0$, $t = 1 - \cos 0 = 0$

when $\theta = \pi$, $t = 1 - \cos \pi = 2$.

$$\text{Thus, } I = \frac{a^2}{2} \int_0^2 t^2 \, dt = \frac{a^2}{2} \left(\frac{t^3}{3} \right)_0^2 = \frac{a^2}{6} (2^3 - 0) = \frac{4a^2}{3}.$$



Change of order of integration:

As already seen, for the double integral with variable limits

$$I = \iint_D f(x, y) \, dA \qquad \text{--- (1)}$$

The limits of integration can be fixed from a rough sketch of the domain of integration. Then (1) can be evaluated as a two-fold iterated integral using either

$$I_D = \int_{x=a}^{x=b} \int_{y=f_1(x)}^{y=f_2(x)} f(x, y) \, dy \, dx \qquad \text{--- (2)}$$

or

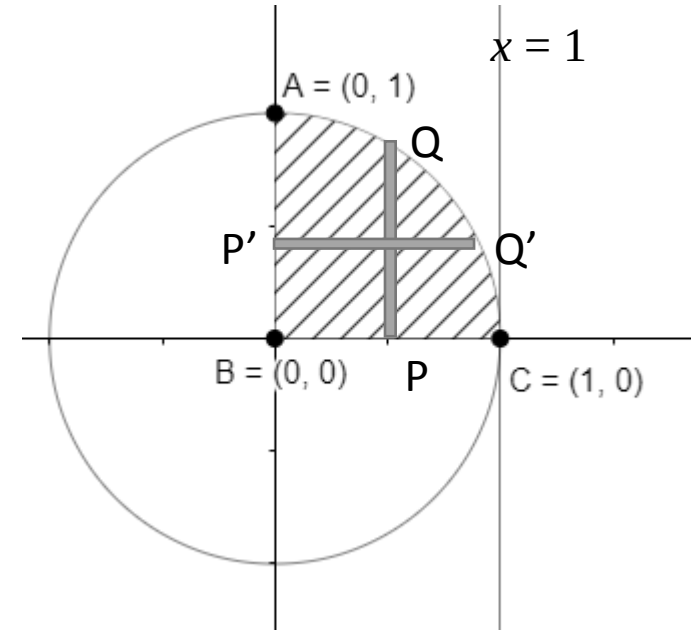
$$I_D = \int_{y=c}^{y=d} \int_{x=f_1(y)}^{x=f_2(y)} f(x, y) \, dx \, dy \qquad \text{--- (3)}$$

In each specific problem, depending upon the type of the domain D and/or the nature of integrand, choose either of the form (2) or (3) whichever is easier to evaluate. Thus in several problems, the evaluation of double integral becomes easier with the change of order of integration, which of course, changes the limits of integration also.

Change of order of integration and hence evaluate the following double integrals:

1. $\int_0^1 \int_0^{\sqrt{1-x^2}} y^2 dy dx$

Solution: From the limits of given double integral, it is clear that first the integration is to be performed w.r.t. y , which varies from $y = 0$ to $y = \sqrt{1-x^2}$, and then w.r.t. x , which varies from $x = 0$ to $x = 1$.



Thus, we have to first integrate along the vertical strip PQ which extends from a point P on the line $y = 0$ (*i.e.*, x - axis) to the point Q on the circle $y = \sqrt{1-x^2}$, (*i.e.*, $x^2 + y^2 = 1$). Then the strip slides from $B(0,0)$ to $C(1,0)$ (*i.e.*, x is varying from 0 to 1).

To change the order of integration consider the horizontal strip $P'Q'$ in the region of integration, which extends from P' on the line $x = 0$ (*i.e.*, y - axis) to the point Q' on the circle

$x = \sqrt{1 - y^2}$ (i.e., $x^2 + y^2 = 1$). Then this strip slides from B to A $(0, 1)$, i.e., y varies from 0 to 1. Therefore,

$$\begin{aligned} I &= \int_0^1 \int_0^{\sqrt{1-y^2}} y^2 \, dy \, dx = \int_0^1 \left\{ \int_0^{\sqrt{1-y^2}} y^2 \, dx \right\} dy = \int_0^1 [y^2 x]_0^{\sqrt{1-y^2}} dy \\ &= \int_0^1 [y^2 \sqrt{1-y^2} - 0] dy = \int_0^1 y^2 \sqrt{1-y^2} dy \end{aligned}$$

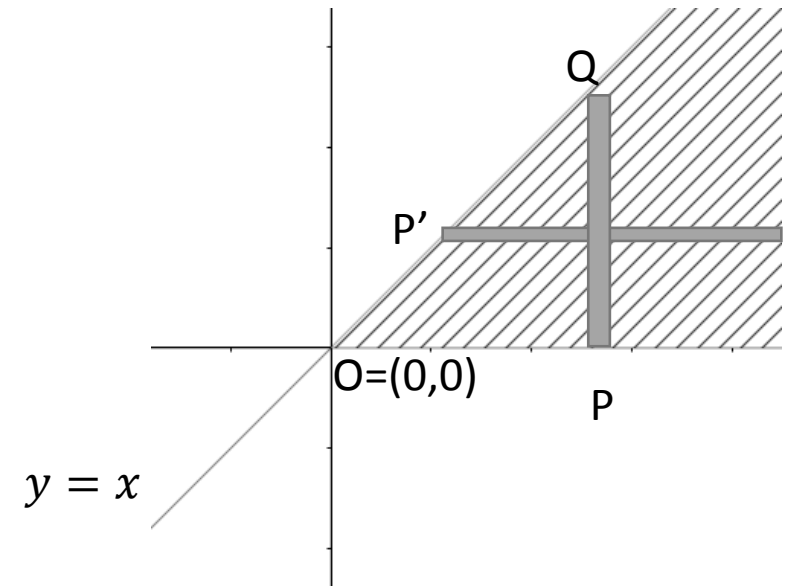
Put $y = \sin \theta$, then $dy = \cos \theta \, d\theta$,

when $y = 0$, $\theta = 0$ and $y = 1$, $\theta = \frac{\pi}{2}$.

$$\begin{aligned} \text{Thus, } I &= \int_0^{\pi/2} \sin^2 \theta \sqrt{1 - \sin^2 \theta} \cos \theta \, d\theta = \int_0^{\pi/2} \sin^2 \theta \cos^3 \theta \, d\theta \\ &= \frac{1 \times 2}{5 \times 3 \times 1} = \frac{2}{15}. \end{aligned}$$

$$2. \int_0^\infty \int_0^x x e^{\frac{-x^2}{y}} dy dx$$

Solution: From the limits of given double integral, it is clear that first the integration is to be performed w.r.t. y , which varies from $y = 0$ to $y = x$, and then w.r.t. x , which varies from $x = 0$ to $x = \infty$.



Thus, we have to first integrate along the vertical strip PQ which extends from a point P on the line $y = 0$ (i.e., x - axis) to the point Q on the line $y = x$. Then the strip slides from $O(0,0)$ to ∞ (i.e., x varies from 0 to ∞).

To change the order of integration consider the horizontal strip $P'Q'$ in the region of integration, which extends from P' on the line $x = y$ to ∞ . Then this strip slides from O to ∞ , i.e., y varies from 0 to ∞ . Therefore,

$$I = \int_0^\infty \int_0^x x e^{\frac{-x^2}{y}} dy dx = \int_0^\infty \left\{ \int_y^\infty x e^{\frac{-x^2}{y}} dx \right\} dy$$

Put $x^2 = t$, then $2x dx = dt$
 When $x = y$, $t = x^2 = y^2$
 When $x = \infty$, $t = \infty$

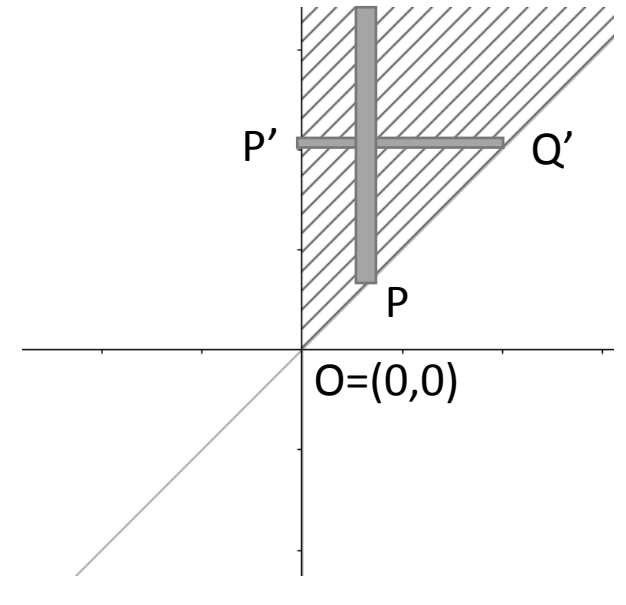
$$\begin{aligned} &= \int_0^\infty \left\{ \int_{y^2}^\infty e^{\frac{-t}{y}} \left(\frac{dt}{2} \right) \right\} dy = \frac{1}{2} \int_0^\infty \left[\frac{e^{\frac{-t}{y}}}{-\frac{1}{y}} \right]_0^\infty dy = \frac{1}{2} \int_0^\infty -y \left(e^{-\infty} - e^{\left(-\frac{y^2}{y}\right)} \right) dy \\ &= \frac{1}{2} \int_0^\infty y e^{-y} dy = \frac{1}{2} \left\{ -y e^{-y} + \int e^{-y} dy \right\}_0^\infty = \frac{1}{2} \left\{ -y e^{-y} - e^{-y} \right\}_0^\infty \end{aligned}$$

Thus,

$$I = \{-y e^{-\infty} - e^{-\infty} + 0 + e^0\} = \frac{1}{2}.$$

$$3. \int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy \, dx$$

Solution: From the limits of given double integral, it is clear that first the integration is to be performed w.r.t. y , which varies from $y = x$ to $y = \infty$, and then w.r.t. x , which varies from $x = 0$ to $x = \infty$.



Thus, we have to first integrate along the vertical strip PQ which extends from a point P on the line $y = x$ to ∞ . Then the strip slides from $O(0, 0)$ to ∞ (*i.e., x varies from 0 to ∞*).

To change the order of integration consider the horizontal strip $P'Q'$ in the region of integration, which extends from P' on the line $x = 0$ (y - axis) to the line $y = x$. Then this strip slides from O to ∞ , i.e., y varies from 0 to ∞ . Therefore,

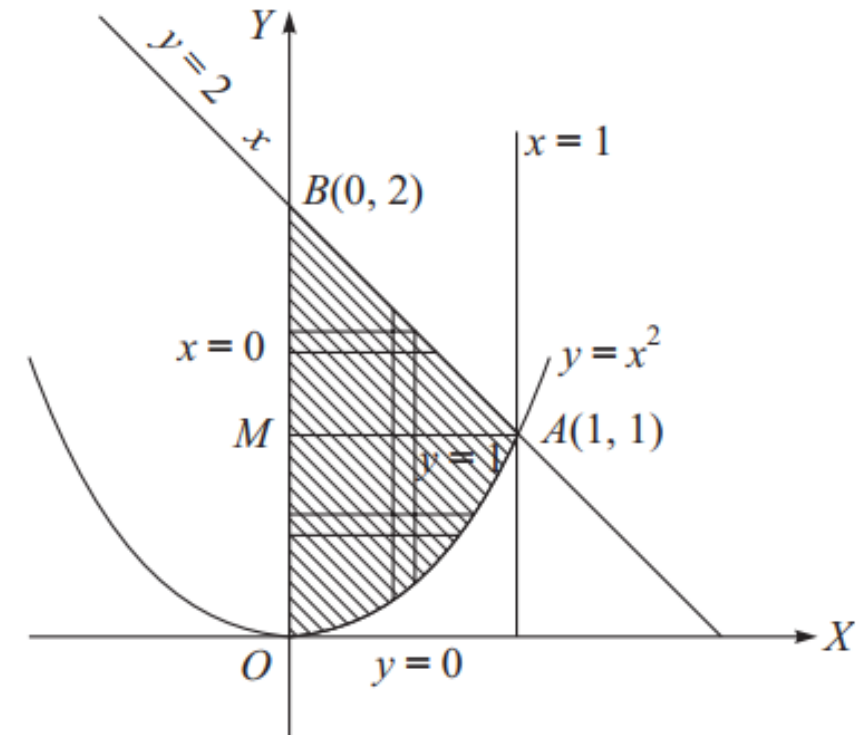
$$I = \int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy \, dx = \int_0^{\infty} \int_0^y \frac{e^{-y}}{y} dx \, dy$$

$$\begin{aligned}
 &= \int_0^{\infty} \left[\frac{e^{-y}}{y} x \right]_0^y dy &= \int_0^{\infty} \left[\frac{e^{-y}}{y} y - \frac{e^{-y}}{y} \cdot 0 \right] dy \\
 &= \int_0^{\infty} e^{-y} dy &= [-e^{-y}]_0^{\infty} = -e^{-\infty} + e^0 = 1.
 \end{aligned}$$

4. $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$

Solution: From the limits of integration, it is clear that the integration is to be performed first w.r.t. y which varies from $y = x^2$ to $y = 2 - x$ and then w.r.t. x which varies from $x = 0$ to $x = 1$. The shaded region in Fig. is the region of integration. Divide this region into vertical strips.

To change the order of integration, divide the region of integration into horizontal strips.



Solving $y = x^2$ to $y = 2 - x$, we get the coordinates of A as $(1, 1)$. Draw $AM \perp OY$. The region of integration is divided into two parts, OAM and MAB .

For the region OAM , x varies from 0 to $\sqrt{y}$ and y varies from 0 to 1. For the region MAB , x varies from 0 to $(2 - y)$ and y varies from 1 to 2. Therefore,

$$\begin{aligned}
 \int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx &= \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy + \int_1^2 \int_0^{2-y} xy \, dx \, dy \\
 &= \int_0^1 y \cdot \left[\frac{x^2}{2} \right]_0^{\sqrt{y}} dy + \int_1^2 y \cdot \left[\frac{x^2}{2} \right]_0^{2-y} dy = \frac{1}{2} \int_0^1 y^2 dy + \frac{1}{2} \int_1^2 y(2-y)^2 dy \\
 &= \frac{1}{2} \cdot \left[\frac{y^3}{3} \right]_0^1 + \frac{1}{2} \int_1^2 (4y - 4y^2 + y^3) dy = \frac{1}{6} + \frac{1}{2} \left[2y^2 - \frac{4y^3}{3} + \frac{y^4}{4} \right]_1^2 \\
 &= \frac{1}{6} + \frac{1}{2} \left[\left(8 - \frac{32}{3} + 4 \right) - \left(2 - \frac{4}{3} + \frac{1}{4} \right) \right] = \frac{3}{8}
 \end{aligned}$$

Change of Variables: Cartesian to Polar Coordinates

For a double integral in cartesian coordinates x, y , the change of variables to polar coordinates r, θ can be done through the transformation $x = r \cos \theta, \quad y = r \sin \theta$ (i.e., $u = r, v = \theta$)

The Jacobian in this case is:

$$J = J \left(\frac{x, y}{r, \theta} \right) = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r(\cos^2 \theta + \sin^2 \theta) = r$$

Thus the double integral in cartesian coordinates x, y gets transformed to double integral in polar coordinates as follows:

$$\iint_D f(x, y) dx dy = \iint_{D^*} f(r \cos \theta, r \sin \theta) |r| dr d\theta$$

Where D^* is the corresponding domain in polar coordinates.

1. Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy dx$ by changing to polar coordinate.

Solution: Here the region of integral is first quadrant of the $xy - plane$, in I quadrant r varies from 0 to ∞ and θ varies from 0 to $\frac{\pi}{2}$. Therefore,

$$\begin{aligned} I &= \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy = \int_{\theta=0}^{\pi/2} \int_{r=0}^\infty e^{-r^2} r dr d\theta \\ &= -\frac{1}{2} \int_0^{\pi/2} \left\{ \int_0^\infty e^{-r^2} (-2r) dr \right\} d\theta = -\frac{1}{2} \int_0^{\pi/2} \left| e^{-r^2} \right|_0^\infty d\theta = \frac{1}{2} \int_0^{\pi/2} d\theta = \frac{\pi}{4}. \end{aligned}$$

Application of double integral

Area of a Plane Region

The area A of a plane regular region (domain) D is given by a two-fold iterated integral

$$\begin{aligned} A &= \iint_D dA = \iint_D dx \, dy \\ &= \int_{x=a}^b \int_{y=f_1(x)}^{f_2(x)} dy \, dx = \int_{y=c}^d \int_{x=f_1(y)}^{f_2(y)} dx \, dy \end{aligned}$$

Area in polar coordinates is given by,

$$A = \iint_{D^*} r \, dr \, d\theta = \int_{\theta=\theta_1}^{\theta_2} \int_{r=f_1(\theta)}^{f_2(\theta)} r \, dr \, d\theta.$$

1. Find the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution:

A = The area of the region bounded by ellipse = 4(area of the ellipse in the first quadrant)

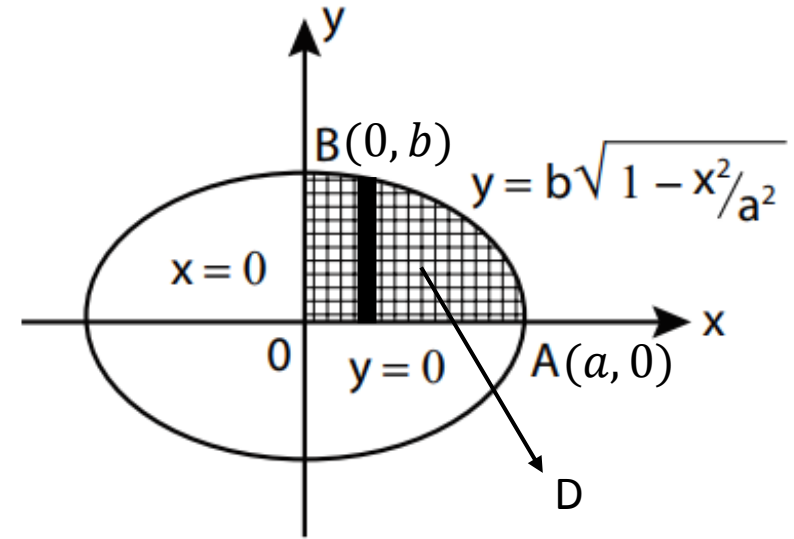
$$A = 4 \iint_D dx \, dy$$

$$= 4 \int_{x=0}^a \int_{y=0}^{\frac{b}{a}\sqrt{a^2-x^2}} dy \, dx$$

$$= 4 \int_{x=0}^a [y]_0^{\frac{b}{a}\sqrt{a^2-x^2}} dx$$

$$= 4 \int_{x=0}^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx$$

$$= \frac{4b}{a} \int_{x=0}^a \sqrt{a^2 - x^2} \, dx$$



put $x = a \sin \theta$, then $dx = a \cos \theta \, d\theta$

when $x = 0$, $\theta = 0$ and $x = a$, $\theta = \frac{\pi}{2}$

$$\text{Therefore, } A = \frac{4b}{a} \int_0^{\pi/2} a^2 \cos^2 \theta \, d\theta = 4ab \times \frac{1}{2} \times \frac{\pi}{2} = \pi ab.$$

2. Find the area the region bounded by the positive y - axis, the arc of the circle $x^2 + y^2 = a^2$ and above the line $y = x$.

Solution:

A = The area of the region D

$$= \iint_D dx \, dy = \int_{x=0}^{a/\sqrt{2}} \int_{y=x}^{\sqrt{a^2-x^2}} dy \, dx$$

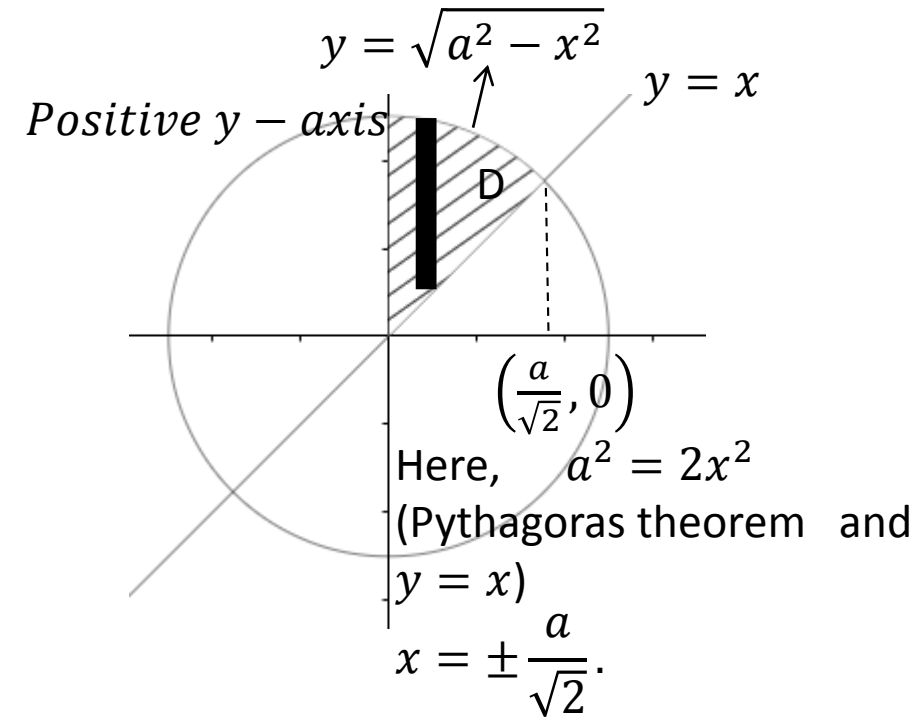
$$= \int_{x=0}^{a/\sqrt{2}} [y]_x^{\sqrt{a^2-x^2}} dx$$

$$= \int_{x=0}^{a/\sqrt{2}} (\sqrt{a^2-x^2} - x) dx = \int_{x=0}^{a/\sqrt{2}} (\sqrt{a^2-x^2}) dx - \int_{x=0}^{a/\sqrt{2}} x dx$$

put $x = a \sin \theta$ in the first term, then $dx = a \cos \theta \, d\theta$

when $x = 0$, $\theta = 0$ and $x = a/\sqrt{2}$, $x = a \sin \theta = a/\sqrt{2}$

$$\text{thus, } \theta = \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4}.$$



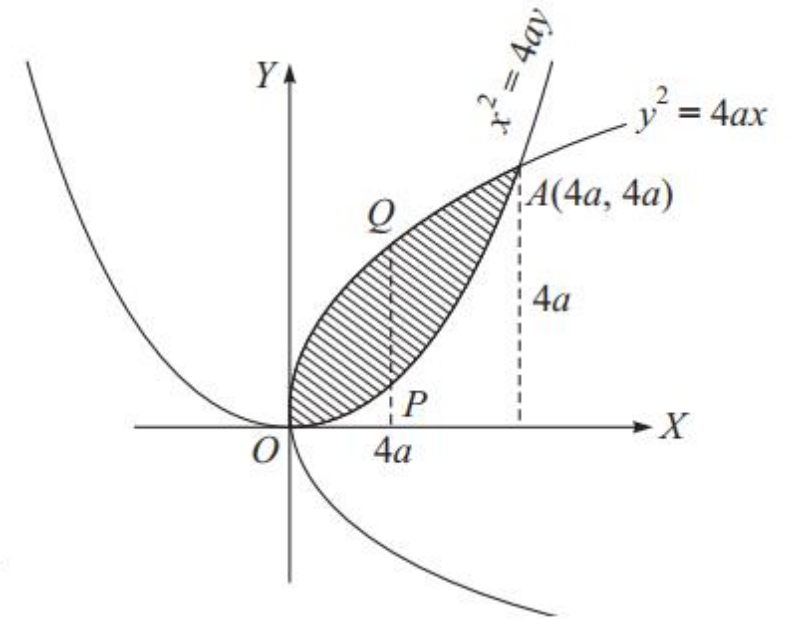
$$\begin{aligned}
\text{Therefore, } A &= \int_0^{\pi/4} a^2 \cos^2 \theta \, d\theta - \int_{x=0}^{\frac{a}{\sqrt{2}}} x \, dx \\
&= \frac{a^2}{2} \int_0^{\pi/4} (1 + \cos 2\theta) \, d\theta - \left(\frac{x^2}{2} \right)_0^{\frac{a}{\sqrt{2}}} \\
&= \frac{a^2}{2} \left\{ \theta + \frac{\sin 2\theta}{2} \right\}_0^{\pi/4} - \frac{a^2}{4} = \frac{a^2}{2} \left\{ \frac{\pi}{4} + \frac{1}{2} \right\} - \frac{a^2}{4} \\
&= \frac{\pi a^2}{8}.
\end{aligned}$$

Note: Here we can also integrate inner integral w. r. t x first, but you need to divide region into 2 sub regions.

3. Show that the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16a^2}{3}$ using double integration.

Solution: Solving the equations $y^2 = 4ax$ and $x^2 = 4ay$, we can see that the parabolas intersect at $O(0, 0)$ and $A(4a, 4a)$.

For the shaded region between these parabolas (shown in figure) x varies from 0 to $4a$ and y varies from P to Q ,



i.e. from $y = \frac{x^2}{4a}$ to $y = 2\sqrt{ax}$. Therefore,

$$\text{Required area} = \int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy \, dx = \int_0^{4a} (2\sqrt{ax} - x^2/4a) dx$$

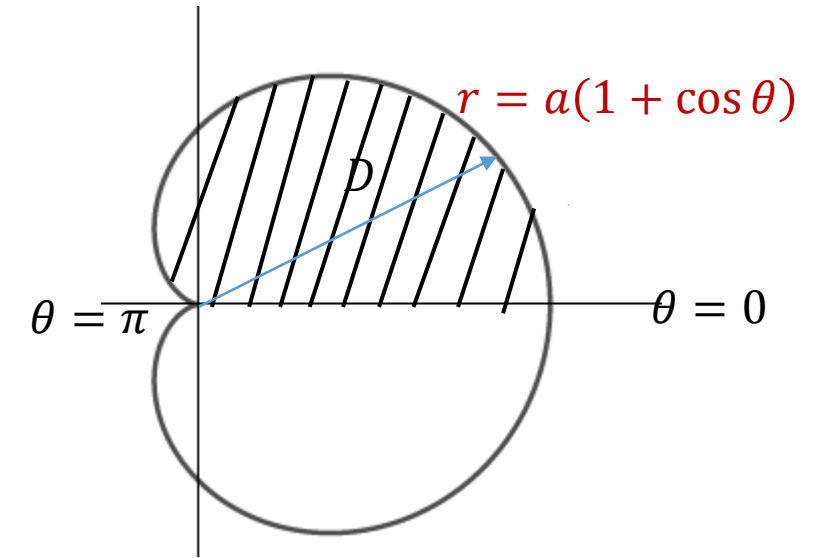
$$= \left[2\sqrt{a} \cdot \frac{2}{3} x^{3/2} - \frac{1}{4a} \cdot \frac{x^3}{3} \right]_0^{4a} = \frac{32}{3} a^2 - \frac{16}{3} a^2 = \frac{16}{3} a^2$$

4. Using double integral find the area by cardioid $r = a(1 + \cos \theta)$.

Solution:

Since cardioid is symmetric about the initial line,

A = Area bounded by the cardioid = 2 (Area of the region D)



$$A = 2 \iint_D dr d\theta = \int_0^\pi \int_0^{a(1+\cos \theta)} r dr d\theta$$

$$= 2 \int_0^\pi \left(\frac{r^2}{2} \right)_0^{a(1+\cos \theta)} d\theta$$

$$= 2 \int_0^\pi \left(\frac{a^2(1+\cos \theta)^2}{2} - 0 \right) d\theta = a^2 \int_0^\pi (1 + \cos \theta)^2 d\theta = a^2 \int_0^\pi \left(2 \cos^2 \left(\frac{\theta}{2} \right) \right)^2 d\theta$$

$$\text{Put } \frac{\theta}{2} = t \text{ then } \frac{d\theta}{2} = dt, \text{ when } \theta = 0, t = 0 \text{ and } \theta = \pi, t = \frac{\pi}{2}.$$

$$\text{Therefore, } A = 8a^2 \int_0^{\pi/2} \cos^4 t dt = 8a^2 \times \frac{3 \times 1}{4 \times 2} \times \frac{\pi}{2} = \frac{3}{2} \pi a^2.$$

Triple integrals:

Triple integral is a generalization of a double integral. Let V be a given three-dimensional domain in space, bounded by a closed surface S . Let $f(x, y, z)$ be a continuous function in V of the rectangular coordinates x, y, z .

Then a triple integral of f over the domain V , denoted by $\iiint_V f(x, y, z) dV$ is defined as

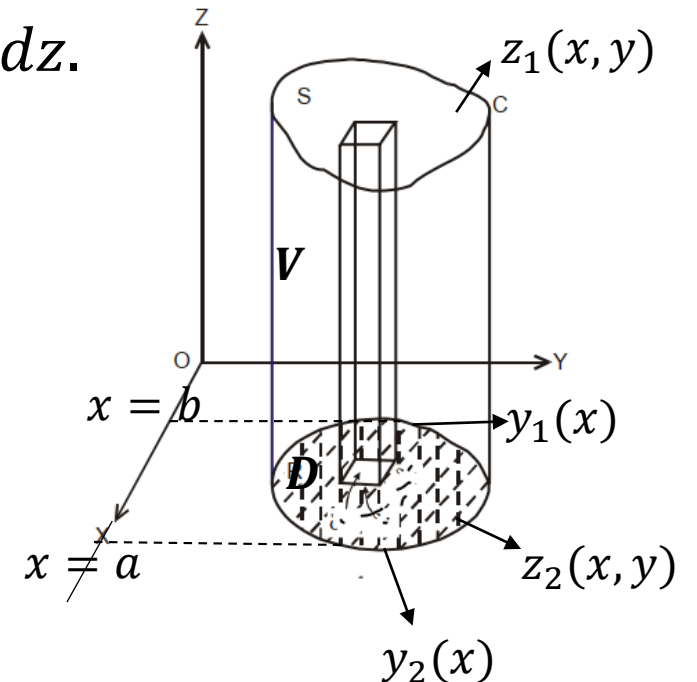
$$\iiint_V f(x, y, z) dV = \iiint_V f(x, y, z) dx dy dz.$$

Evaluation of a Triple Integral

Let the equations of the surfaces bounding a regular domain V below and above be $z = z_1(x, y)$ and $z = z_2(x, y)$ respectively.

Let the projection D of V onto xy -plane be bounded by the $y = y_1(x)$ and $y = y_2(x)$ and $x = a, x = b$.

Then the three fold integral I of a continuous function $f(x, y, z)$ over a regular domain V is



defined as

$$I = \int_{x=a}^{x=b} \left[\int_{y=y_1(x)}^{y=y_2(x)} \left\{ \int_{z=z_1(x,y)}^{z=z_2(x,y)} f(x, y, z) dz \right\} dy \right] dx.$$

Here the limits of integration are chosen to cover the domain V by varying z from lower surface $z = z_1(x, y)$ to the upper surface $z = z_2(x, y)$ and covering the projection D by varying y from $y_1(x)$ to $y_2(x)$ and x from a to b .

In the three-fold iterated integral, inner integration is done first with respect to z (i.e., within the curly brackets) with the substitution of limits for z , the next integration is carried with respect to y (i.e., within the square brackets). This results in an integrand which is a function of x alone which is then integrated w.r.t. x between a and b .

Note: When V is projected on to xz –plane or yz – plane instead of xy –plane, then the order of integration and the limits are to be rewritten appropriately.

Evaluate the following triple integrals:

1. $\int_0^2 \int_1^z \int_0^{yz} xyz \, dx \, dy \, dz$

Solution: $I = \int_0^2 \left\{ \int_1^z \left\{ \int_0^{yz} xyz \, dx \right\} dy \right\} dz$

$$= \int_0^2 \left\{ \int_1^z \left\{ \frac{x^2 yz}{2} \right\}_0^{yz} dy \right\} dz = \int_0^2 \left\{ \int_1^z \left\{ \frac{(yz)^2 yz}{2} \right\} dy \right\} dz$$

$$= \frac{1}{2} \int_0^2 \left\{ \int_1^z \{y^3 z^3\} dy \right\} dz = \frac{1}{2} \int_0^2 \left\{ \frac{y^4}{4} z^3 \right\}_1^z dz = \frac{1}{8} \int_0^2 (z^7 - z^3) dz$$

$$= \frac{1}{8} \left(\frac{z^8}{8} - \frac{z^4}{4} \right)_0^2 = \frac{1}{8} \left(\frac{2^8}{8} - \frac{2^4}{4} \right)$$

$$= \frac{7}{2}.$$

$$2. \int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dx dy dz$$

Solution:

$$\begin{aligned}
 I &= \int_0^a \left\{ \int_0^x \left\{ \int_0^{x+y} e^{(x+y)+z} dz \right\} dy \right\} dx = \int_0^a \left\{ \int_0^x \left\{ \int_0^{x+y} e^{(x+y)} e^z dz \right\} dy \right\} dx \\
 &= \int_0^a \left\{ \int_0^x \left\{ e^{(x+y)} e^z \right\}_0^{x+y} dy \right\} dx = \int_0^a \left\{ \int_0^x \left\{ e^{(x+y)} e^{x+y} - e^{x+y} e^0 \right\} dy \right\} dx \\
 &= \int_0^a \left\{ \int_0^x \left\{ e^{2x+2y} - e^{x+y} \right\} dy \right\} dx = \int_0^a \left\{ \int_0^x \left\{ e^{2x} e^{2y} - e^x e^y \right\} dy \right\} dx \\
 &= \int_0^a \left\{ \left[e^{2x} \frac{e^{2y}}{2} - e^x e^y \right]_0^x \right\} dx = \int_0^a \left\{ \left[e^{2x} \frac{e^{2x}}{2} - e^x e^x - e^{2x} \frac{e^0}{2} + e^x e^0 \right] \right\} dx \\
 &= \int_0^a \left\{ \left[\frac{e^{4x}}{2} - \frac{3e^{2x}}{2} + e^x \right] \right\} dx = \left[\frac{e^{4x}}{8} - \frac{3e^{2x}}{4} + e^x \right]_0^a \\
 &= \left[\frac{e^{4a}}{8} - \frac{3e^{2a}}{4} + e^a - \frac{e^0}{8} + \frac{3e^0}{4} - e^0 \right] \\
 &= \frac{e^{4a}}{8} - \frac{3e^{2a}}{4} + e^a - \frac{3}{8}.
 \end{aligned}$$

3. $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) \, dy \, dx \, dz$

Solution: Integrating first with respect to y , keeping x and z constant

$$\begin{aligned} I &= \int_{-1}^1 \int_0^z \left(\left[xy + \frac{y^2}{2} + yz \right]_{x-z}^{x+z} \right) dx \, dz \\ &= \int_{-1}^1 \left(\int_0^z (4zx + 2z^2) \, dx \right) dz \\ &= \int_{-1}^1 \left[4z \frac{x^2}{2} + 2 \cdot z^2 \cdot x \right]_0^z dz \\ &= \int_{-1}^1 \left[4z \cdot \frac{z^2}{2} + 2z^2 \cdot z \right] dz \\ &= 4 \int_{-1}^1 z^3 \, dz = 4 \frac{z^4}{4} \Big|_{-1}^1 = 0 \end{aligned}$$

Applications of Triple Integrals

Volume:

Volume of a solid contained in the domain V is given by the triple integral with $f(x, y, z) = 1$

i.e., $\text{volume} = \iiint_V dx dy dz$.

1. Find the volume bounded by the, $z = x^2 + y^2$, $x = -a$, $x = a$, $y = -a$ and $y = a$.

Solution:

$$\begin{aligned}\text{Volume} &= \int_{-a}^a \left\{ \int_{-a}^a \left\{ \int_0^{x^2+y^2} dz \right\} dy \right\} dx \\ &= \int_{-a}^a \left\{ \int_{-a}^a \{x^2 + y^2\} dy \right\} dx = \int_{-a}^a \left\{ \frac{x^2 y}{2} + \frac{y^3}{3} \right\}_{-a}^a dx \\ &= \int_{-a}^a \left\{ \frac{2x^2 a}{2} + \frac{2a^3}{3} \right\} dx = \left(\frac{ax^3}{3} + \frac{2a^3 x}{3} \right)_{-a}^a = 2a^4.\end{aligned}$$

Beta and Gamma functions:

The beta function, denoted by $\beta(m, n)$, is defined as

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx, \quad m > 0 \text{ and } n > 0.$$

Properties:

1. $\beta(m, n) = \beta(n, m)$

Proof: By the definition of beta function,

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx,$$

Put $x = 1 - y$, then $dx = -dy$. When $x = 0$, $y = 1$ and when $x = 1$, $y = 0$

$$\begin{aligned} \text{Thus we have } \beta(m, n) &= \int_0^1 x^{m-1} (1-x)^{n-1} dx = \int_1^0 (1-y)^{m-1} y^{n-1} (-dy) \\ &= \int_0^1 y^{n-1} (1-y)^{m-1} dy = \beta(n, m). \end{aligned}$$

2. Beta function in terms of trigonometric function

$$\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n} \theta d\theta ,$$

Proof: By the definition of beta function,

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx,$$

Put $x = \sin^2 \theta$, then $dx = 2 \sin \theta \cos \theta d\theta$.

When $x = 0$, $\theta = 0$ and when $x = 1$, $\theta = \frac{\pi}{2}$

$$\begin{aligned} \text{Thus we have } \beta(m, n) &= \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta. \end{aligned}$$

Gamma functions:

The gamma function, denoted by $\Gamma(n)$, is defined as

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx, \quad n > 0.$$

In particular, $\Gamma(1) = \int_0^{\infty} e^{-x} x^{1-1} dx = \int_0^{\infty} e^{-x} dx = (-e^{-x})_0^{\infty} = -e^{\infty} + e^0 = 1$.

Using the technique of integration by parts we have the following very useful property of gamma function.

Reduction formula for gamma function :

$$\Gamma(n + 1) = n\Gamma(n)$$

Proof: By definition,
$$\begin{aligned} \Gamma(n + 1) &= \int_0^{\infty} e^{-x} x^n dx = \left[\frac{x^n e^{-x}}{-1} \right]_0^{\infty} - \left[\int \frac{e^{-x}}{-1} nx^{n-1} dx \right]_0^{\infty} \\ &= 0 + n \int_0^{\infty} e^{-x} x^{n-1} dx = n\Gamma(n). \end{aligned}$$

Note: If n is a positive integer then $\Gamma(n + 1) = n!$.

Relation between beta and gamma functions:

$$\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

Proof: We have $\Gamma(m) = \int_0^\infty e^{-t} t^{m-1} dt$

Put $t = x^2$, then $dt = 2x dx$. When $t = 0$, $x = 0$ and $t = \infty$, $x = \infty$.

$$\text{Thus,} \quad \Gamma(m) = 2 \int_0^\infty e^{-x^2} x^{2m-1} dx \quad \text{----- (1)}$$

$$\text{Similarly,} \quad \Gamma(n) = 2 \int_0^\infty e^{-y^2} y^{2n-1} dy$$

$$\therefore \Gamma(m) \Gamma(n) = 2 \int_0^\infty e^{-x^2} x^{2m-1} dx \times 2 \int_0^\infty e^{-y^2} y^{2n-1} dy$$

$$= \int_0^\infty \int_0^\infty e^{-x^2} x^{2m-1} e^{-y^2} y^{2n-1} dx dy \quad \text{(limits of integrations are constants).}$$

$$= 4 \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2m-1} y^{2n-1} dx dy \quad \text{--- (2)}$$

Now change to polar coordinates by substituting $x = r \cos \theta$, $y = r \sin \theta$ and $dx dy = r dr d\theta$.

To cover the region of integration in (2) which is the entire I quadrant,

r varies from 0 to ∞ and θ varies from 0 to $\frac{\pi}{2}$.

Thus (2) becomes,

$$\begin{aligned} \Gamma(m) \Gamma(n) &= 4 \int_0^{\pi/2} \int_0^\infty e^{-r^2} (r \sin \theta)^{2m-1} (r \cos \theta)^{2n-1} r dr d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \times 2 \int_0^\infty e^{-r^2} r^{2(m+n)-1} dr \end{aligned}$$

But by (1), $2 \int_0^\infty e^{-r^2} r^{2(m+n)-1} dr = \Gamma(m+n)$

and $2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta = \beta(m,n)$

Thus, $\Gamma(m) \Gamma(n) = \Gamma(m+n) \beta(m,n).$

Note: 1. We have $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

put $2m - 1 = p$, then $m = \frac{p+1}{2}$ and put $2n - 1 = q$, then $n = \frac{q+1}{2}$.

Thus, $\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$.

2. Put $m = n = \frac{1}{2}$ in the relation $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$,

then, $\frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}+\frac{1}{2}\right)} = \beta\left(\frac{1}{2}, \frac{1}{2}\right) = 2 \int_0^{\pi/2} \sin^{2\left(\frac{1}{2}\right)-1} \theta \cos^{2\left(\frac{1}{2}\right)-1} \theta d\theta$

$$\left\{\Gamma\left(\frac{1}{2}\right)\right\}^2 = 2 \int_0^{\pi/2} \sin^0 \theta \cos^0 \theta d\theta = 2 \int_0^{\pi/2} d\theta = 2(\theta)_0^{\pi/2} = \pi$$

$$\therefore \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Note: 3. We know that $\Gamma(n) = 2 \int_0^{\infty} e^{-x^2} x^{2n-1} dx$

Put $n = \frac{1}{2}$ in the above equation,

$$\Gamma\left(\frac{1}{2}\right) = 2 \int_0^{\infty} e^{-x^2} x^0 dx = 2 \int_0^{\infty} e^{-x^2} dx$$

Thus,

$$2 \int_0^{\infty} e^{-x^2} dx = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Therefore,

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

1. Find the following.

i) $\Gamma(5/2)$ ii) $\Gamma(7/2)$

Solution:

$$\text{i) } \Gamma\left(\frac{5}{2}\right) = \Gamma\left(\frac{3}{2} + 1\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2}\Gamma\left(\frac{1}{2} + 1\right) = \frac{3}{2} \cdot \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{3\sqrt{\pi}}{4}$$

$$\text{ii) } \Gamma\left(\frac{7}{2}\right) = \Gamma\left(\frac{5}{2} + 1\right) = \frac{5}{2}\Gamma\left(\frac{5}{2}\right) = \frac{5}{2} \cdot \frac{3\sqrt{\pi}}{4} = \frac{15\sqrt{\pi}}{8}$$

2. Prove that $\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \int_0^{\pi/2} \frac{1}{\sqrt{\sin \theta}} d\theta = \pi$.

Solution: Consider,

$$\int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \int_0^{\pi/2} \sin^{1/2} \theta \cos^0 \theta d\theta = \frac{1}{2} \beta\left(\frac{3}{4}, \frac{1}{2}\right) = \frac{1}{2} \frac{\Gamma(\frac{3}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{4} + \frac{1}{2})} = \frac{1}{2} \frac{\Gamma(\frac{3}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{5}{4})} \quad \text{-----(1)}$$

Now consider,

$$\int_0^{\pi/2} \frac{1}{\sqrt{\sin \theta}} d\theta = \int_0^{\pi/2} \sin^{-\frac{1}{2}} \theta \cos^0 \theta d\theta = \frac{1}{2} \beta\left(\frac{1}{4}, \frac{1}{2}\right) = \frac{1}{2} \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{1}{4} + \frac{1}{2})} = \frac{1}{2} \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{4})} \quad \text{-----(2)}$$

From (1) and (2),

$$\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \cdot \int_0^{\pi/2} \frac{1}{\sqrt{\sin \theta}} d\theta = \frac{1}{2} \frac{\Gamma(\frac{3}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{5}{4})} \cdot \frac{1}{2} \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{4})}$$

$$\begin{aligned}
&= \frac{1}{4} \frac{\Gamma(\frac{1}{4}) [\Gamma(\frac{1}{2})]^2}{\Gamma(\frac{1}{4} + 1)} \\
&= \frac{\Gamma(\frac{1}{4}) [\Gamma(\frac{1}{2})]^2}{\Gamma(\frac{1}{4})} \quad (\text{as } \Gamma(n + 1) = n\Gamma(n)) \\
&= \pi \quad (\text{as } \Gamma(1/2) = \sqrt{\pi})
\end{aligned}$$

3. Express the following in terms of gamma function

$$\text{i) } \int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta \quad \text{ii) } \int_0^{\infty} \frac{1}{1+x^4} \, dx \quad \text{iii) } \int_0^2 x (8-x^3)^{1/3} \, dx$$

Solution:

$$\text{i) Let } I = \int_0^{\pi/2} \sqrt{\tan \theta} \, d\theta. \text{ Since } \tan \theta = \frac{\sin \theta}{\cos \theta}, \text{ we get}$$

$$I = \int_0^{\pi/2} \sin^{1/2} \theta \cos^{-1/2} \theta d\theta = \frac{1}{2} \beta \left(\frac{3}{4}, \frac{1}{4} \right) = \frac{1}{2} \frac{\Gamma \left(\frac{3}{4} \right) \Gamma \left(\frac{1}{4} \right)}{\Gamma \left(\frac{3}{4} + \frac{1}{4} \right)} = \frac{1}{2} \Gamma \left(\frac{3}{4} \right) \Gamma \left(\frac{1}{4} \right)$$

ii) Let $I = \int_0^{\infty} \frac{1}{1+x^4} dx$. Putting $x^2 = \tan \theta$ we get

$$I = \int_0^{\pi/2} \frac{1}{\sec^2 \theta} \frac{\sec^2 \theta}{2\sqrt{\tan \theta}} d\theta = \frac{1}{2} \int_0^{\pi/2} \sin^{-1/2} \theta \cos^{1/2} \theta d\theta = \frac{1}{4} \beta \left(\frac{1}{4}, \frac{3}{4} \right) = \frac{1}{4} \frac{\Gamma \left(\frac{1}{4} \right) \Gamma \left(\frac{3}{4} \right)}{\Gamma \left(\frac{1}{4} + \frac{3}{4} \right)} = \frac{1}{4} \Gamma \left(\frac{1}{4} \right) \Gamma \left(\frac{3}{4} \right)$$

iii) Let $I = \int_0^2 x (8-x^3)^{1/3} dx = 2 \int_0^2 x \left(1 - \frac{x^3}{8} \right)^{1/3} dx$. Put $t = 1 - \frac{x^3}{8}$ i.e., $x = 2(1-t)^{1/3}$.

This implies $dx = -\frac{2}{3}(1-t)^{-2/3} dt$. When $x = 0$, $t = 1$ and when $x = 2$, $t = 0$. Therefore

$$I = \frac{8}{3} \int_0^1 t^{1/3} (1-t)^{-1/3} dt = \frac{8}{3} \beta \left(\frac{4}{3}, \frac{2}{3} \right) = \frac{8}{3} \frac{\Gamma \left(\frac{4}{3} \right) \Gamma \left(\frac{2}{3} \right)}{\Gamma(2)} = \frac{8}{3} \Gamma \left(\frac{4}{3} \right) \Gamma \left(\frac{2}{3} \right)$$

4. Evaluate $\int_0^1 x^7(1-x^4)^3 dx$ in terms of gamma function.

Solution. Let $I = \int_0^1 x^7(1-x^4)^3 dx$. Put $t = 1 - x^4$. This implies $-4x^3 dx = dt$. When $x = 0$, $t = 1$ and when $x = 1$, $t = 0$. Therefore we have

$$I = -\frac{1}{4} \int_1^0 (1-t) t^3 dt = \frac{1}{4} \int_0^1 t^3 (1-t) dt = \frac{1}{4} \beta(4, 2) = \frac{1}{4} \frac{\Gamma(4)\Gamma(2)}{\Gamma(4+2)} = \frac{1}{4} \frac{\Gamma(4)\Gamma(2)}{\Gamma(6)} = \frac{1}{4} \frac{3! 1!}{5!} = \frac{1}{80}$$

5. Evaluate the followings in terms of gamma function.s

$$\text{i) } \int_0^{\pi/2} \sin^3 \theta \cos^{-\frac{1}{2}} \theta d\theta \quad \text{ii) } \int_0^{\pi/2} \sin^4 \theta \cos^6 \theta d\theta \quad \text{iii) } \int_0^{\pi/2} \sin^9 \theta \cos^5 \theta d\theta \quad \text{iv) } \int_0^{\pi/2} \sin^2 \theta \cos^5 \theta d\theta$$

NOTE:1. Refer the class notes for extra problems.

2. Don't blindly follow the answers, there may be some typographical errors, so try to solve each problem by your self.